

Class08: Breast Cancer Mini Project

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Table of contents

Background	1
Data Import	1
Exploratory Data Analysis	2
Principal Component Analysis (PCA)	3
Variance explained	6
Hierarchical Clustering	8
Combining Methods	9

Background

The goal of this mini-project is for you to explore a complete analysis using the unsupervised learning techniques covered in class.

Today we will analyze a biopsy dataset from fine needle aspiration (FNA) of a breast mass.

Data Import

The data is made available as a CSV file for download. We can read this using ‘read.csv()’:

```
wisc.df <- read.csv("WisconsinCancer.csv")
```

```
head(wisc.df, 3)
```

```
      id diagnosis radius_mean texture_mean perimeter_mean area_mean
1  842302         M    17.99      10.38         122.8        1001
2  842517         M    20.57      17.77         132.9        1326
3 84300903         M    19.69      21.25         130.0        1203
 smoothness_mean compactness_mean concavity_mean concave.points_mean
```

1	0.11840	0.27760	0.3001	0.14710	
2	0.08474	0.07864	0.0869	0.07017	
3	0.10960	0.15990	0.1974	0.12790	
	symmetry_mean	fractal_dimension_mean	radius_se	texture_se	perimeter_se
1	0.2419	0.07871	1.0950	0.9053	8.589
2	0.1812	0.05667	0.5435	0.7339	3.398
3	0.2069	0.05999	0.7456	0.7869	4.585
	area_se	smoothness_se	compactness_se	concavity_se	concave.points_se
1	153.40	0.006399	0.04904	0.05373	0.01587
2	74.08	0.005225	0.01308	0.01860	0.01340
3	94.03	0.006150	0.04006	0.03832	0.02058
	symmetry_se	fractal_dimension_se	radius_worst	texture_worst	perimeter_worst
1	0.03003	0.006193	25.38	17.33	184.6
2	0.01389	0.003532	24.99	23.41	158.8
3	0.02250	0.004571	23.57	25.53	152.5
	area_worst	smoothness_worst	compactness_worst	concavity_worst	
1	2019	0.1622	0.6656	0.7119	
2	1956	0.1238	0.1866	0.2416	
3	1709	0.1444	0.4245	0.4504	
	concave.points_worst	symmetry_worst	fractal_dimension_worst		
1	0.2654	0.4601	0.11890		
2	0.1860	0.2750	0.08902		
3	0.2430	0.3613	0.08758		

Make sure we remove or exclude the diagnosis column from the dataset that we use for further analysis - this is the expert diagnosis as either M or B.

```
wisc.data <- wisc.df[,-c(1,2)]
diagnosis <- as.factor(wisc.df$diagnosis)
```

Exploratory Data Analysis

Q1. How many observations are in this dataset?

```
nrow(wisc.data)
```

```
[1] 569
```

Q2. How many of the observations have a malignant diagnosis?

```
table(wisc.df$diagnosis)
```

```
  B    M  
357 212
```

Q3. How many variables/features in the data are suffixed with ' _mean'?

We can use 'grep()' function to help us here:

```
colnames(wisc.data)
```

```
[1] "radius_mean"      "texture_mean"  
[3] "perimeter_mean"   "area_mean"  
[5] "smoothness_mean"  "compactness_mean"  
[7] "concavity_mean"    "concave.points_mean"  
[9] "symmetry_mean"     "fractal_dimension_mean"  
[11] "radius_se"         "texture_se"  
[13] "perimeter_se"      "area_se"  
[15] "smoothness_se"     "compactness_se"  
[17] "concavity_se"      "concave.points_se"  
[19] "symmetry_se"       "fractal_dimension_se"  
[21] "radius_worst"      "texture_worst"  
[23] "perimeter_worst"   "area_worst"  
[25] "smoothness_worst"  "compactness_worst"  
[27] "concavity_worst"   "concave.points_worst"  
[29] "symmetry_worst"    "fractal_dimension_worst"
```

```
length(grep("_mean", colnames(wisc.data), value = T))
```

```
[1] 10
```

Principal Component Analysis (PCA)

We need to scale out data before PCA with the 'scale=TRUE' argument to 'prcomp()'.

```
wisc.pr <- prcomp(wisc.data, scale = TRUE)  
summary(wisc.pr)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	3.6444	2.3857	1.67867	1.40735	1.28403	1.09880	0.82172
Proportion of Variance	0.4427	0.1897	0.09393	0.06602	0.05496	0.04025	0.02251
Cumulative Proportion	0.4427	0.6324	0.72636	0.79239	0.84734	0.88759	0.91010
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
Standard deviation	0.69037	0.6457	0.59219	0.5421	0.51104	0.49128	0.39624
Proportion of Variance	0.01589	0.0139	0.01169	0.0098	0.00871	0.00805	0.00523
Cumulative Proportion	0.92598	0.9399	0.95157	0.9614	0.97007	0.97812	0.98335
	PC15	PC16	PC17	PC18	PC19	PC20	PC21
Standard deviation	0.30681	0.28260	0.24372	0.22939	0.22244	0.17652	0.1731
Proportion of Variance	0.00314	0.00266	0.00198	0.00175	0.00165	0.00104	0.0010
Cumulative Proportion	0.98649	0.98915	0.99113	0.99288	0.99453	0.99557	0.9966
	PC22	PC23	PC24	PC25	PC26	PC27	PC28
Standard deviation	0.16565	0.15602	0.1344	0.12442	0.09043	0.08307	0.03987
Proportion of Variance	0.00091	0.00081	0.0006	0.00052	0.00027	0.00023	0.00005
Cumulative Proportion	0.99749	0.99830	0.9989	0.99942	0.99969	0.99992	0.99997
	PC29	PC30					
Standard deviation	0.02736	0.01153					
Proportion of Variance	0.00002	0.00000					
Cumulative Proportion	1.00000	1.00000					

Q4. From your results, what proportion of the original variance is captured by the first principal component (PC1)?

0.4427, The first principal component (PC1) captures approximately 44.27% of the total variance in the data.

Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

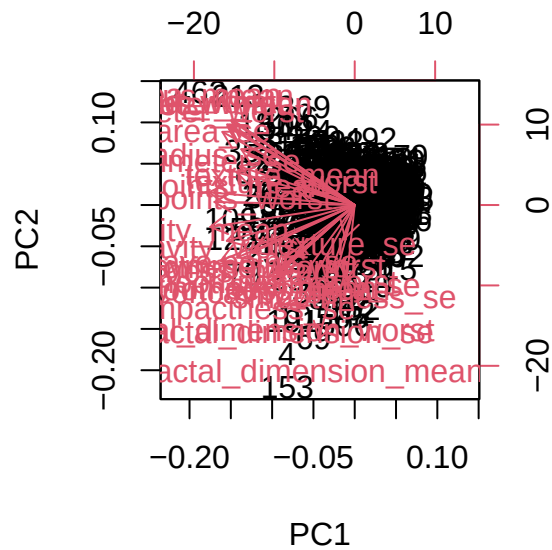
At least 70% variance is explained by the first 3 principal components.

Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

At least 90% variance is explained by the first 7 principal components.

Let's see the biplot

```
biplot(wisc.pr)
```



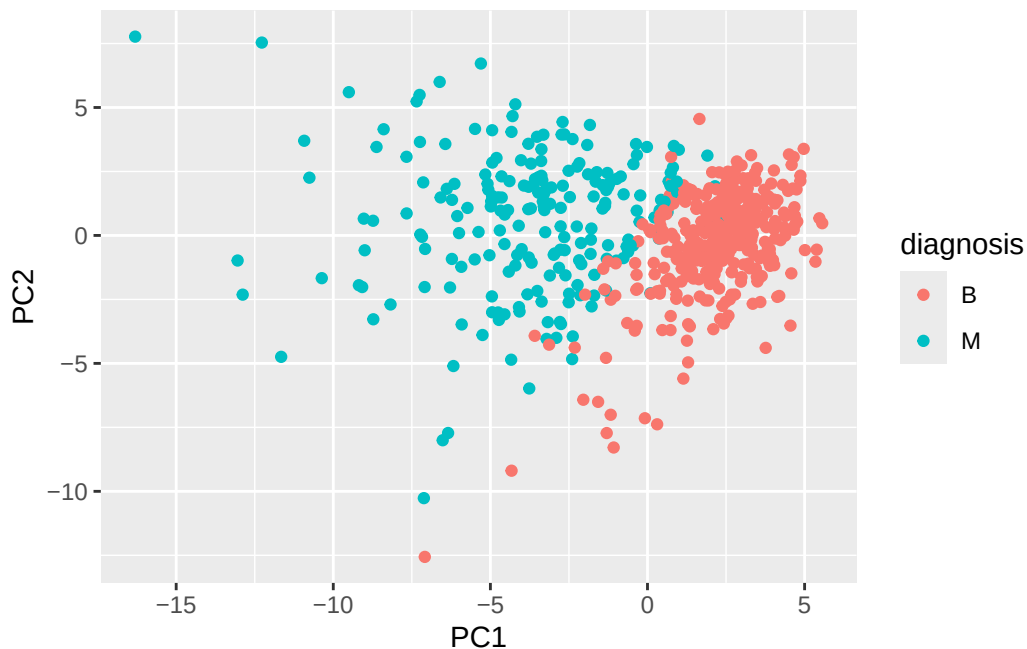
Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

The biplot is difficult to interpret because it is very cluttered, with many overlapping labels for both observations and variables. This makes it hard to clearly see patterns or groupings in the data. While it contains a lot of information, the visual complexity makes it challenging to extract meaningful insights compared to simpler PCA plots like PC1 vs PC2.

Let's see the 'PC score plot'

```
library(ggplot2)

ggplot(wisc.pr$x) +
  aes(PC1, PC2, col=diagnosis) +
  geom_point()
```



Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

The PC1 vs PC3 plot shows less clear separation between malignant and benign samples compared to the PC1 vs PC2 plot. While PC1 still captures the main variation and helps distinguish the groups, PC3 does not add much additional separation. This suggests that PC2 is more informative than PC3 for distinguishing between the two diagnoses, and most of the important structure in the data is captured by PC1 and PC2.

Variance explained

A scree plot shows how much variance each PC captures. We typically look for an “elbow” — a point where adding more PCs gives diminishing returns. This can help us decide how many PCs to consider for further analysis.

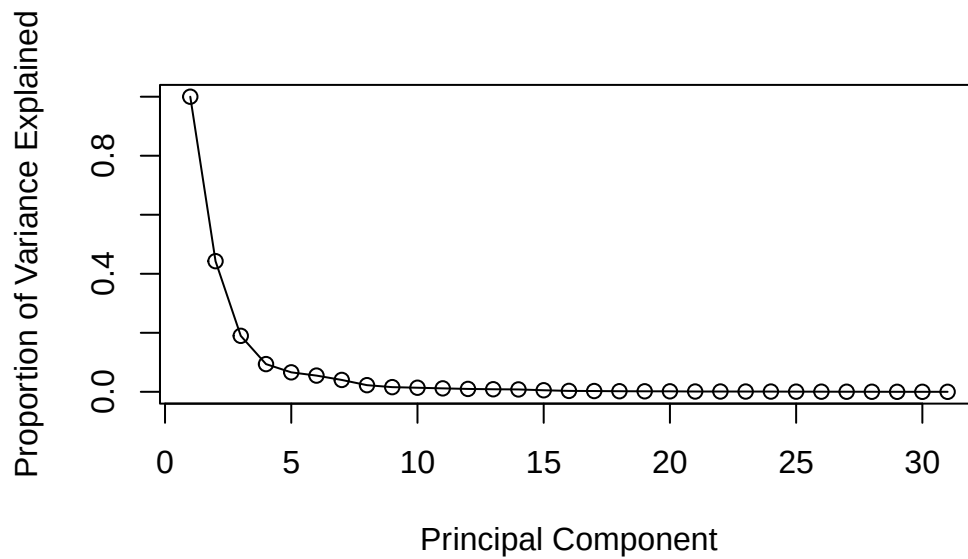
Calculate variance of each component

```
pr.var <- wisc.pr$sdev^2
head(pr.var)
```

```
[1] 13.281608  5.691355  2.817949  1.980640  1.648731  1.207357
```

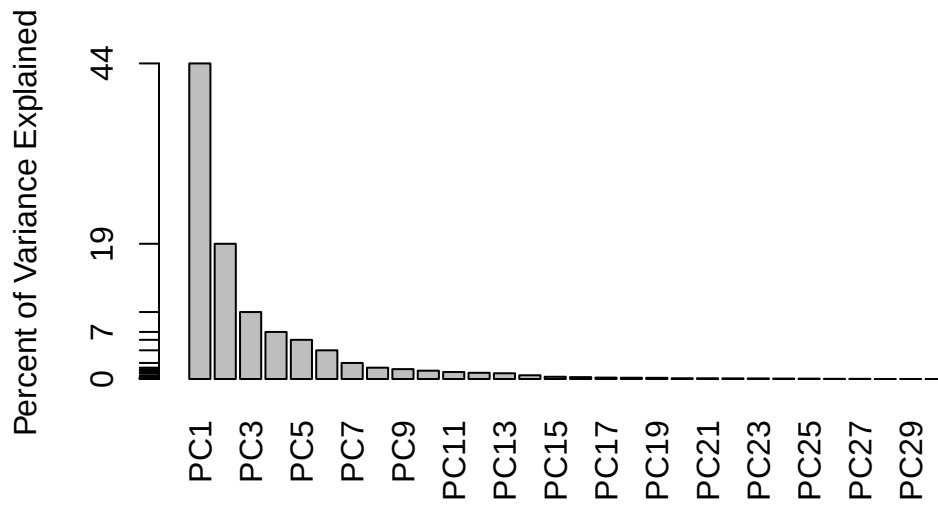
```
pve <- pr.var / sum(pr.var)

plot(c(1, pve),
     xlab = "Principal Component",
     ylab = "Proportion of Variance Explained",
     ylim = c(0, 1),
     type = "o")
```



```
barplot(pve,
        ylab = "Percent of Variance Explained",
        names.arg = paste0("PC", 1:length(pve)),
        las = 2,
        axes = FALSE)

axis(2, at = pve, labels = round(pve, 2) * 100)
```



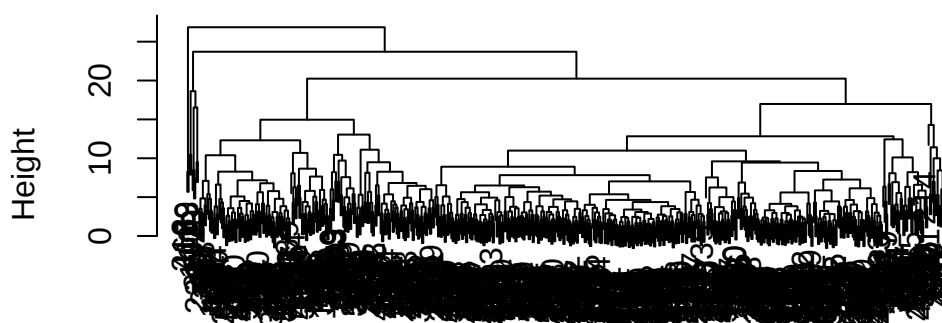
Q9. For the first principal component, what is the component of the loading vector (i.e. `wisc.pr$rotation[,1]`) for the feature `concave.points_mean`? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

The loading of `concave.points_mean` on the first principal component is approximately 0.26, indicating it contributes strongly to PC1. Yes, there are a few features with slightly larger contributions (such as `radius_mean`, `perimeter_mean`, and `area_mean`), but `concave.points_mean` is among the most influential features driving variation in PC1.

Hierarchical Clustering

```
wisc.hclust <- hclust( dist( scale(wisc.data)))
plot(wisc.hclust)
```


Cluster Dendrogram



```
dist(scale(wisc.data))  
hclust(*, "complete")
```

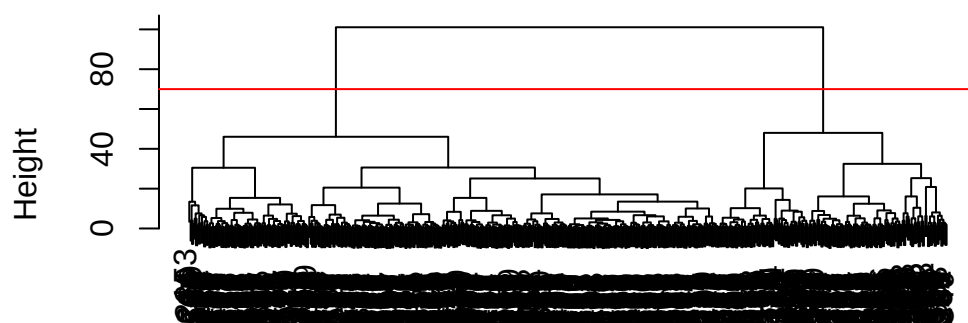
Q10. Using the `plot()` and `abline()` functions, what is the height at which the clustering model has 4 clusters?

The clustering model has approximately 4 clusters at a height around 15–18, based on where a horizontal cut through the dendrogram separates the data into four main groups.

Combining Methods

```
d <- dist(wisc.pr$x[,1:4])  
wisc.pr.hclust <- hclust(d, method = "ward.D2")  
plot(wisc.pr.hclust)  
abline(h=70, col="red")
```

Cluster Dendrogram



d
hclust (*, "ward.D2")

```
grps <- cutree(wisc.pr.hclust, h=70)
table(grps)
```

```
grps
  1  2
171 398
```

How does this clustering 'grps' correspond to the diagnosis.

```
table(diagnosis)
```

```
diagnosis
  B  M
357 212
```

```
table(diagnosis, grps)
```

```
      grps
diagnosis  1  2
  B      6 351
  M     165  47
```